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A head-to-head *NLR* gene pair from wild emmer confers stripe rust resistance in wheat

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Wheat stripe rust poses a major threat to global food security. Discovery of disease resistance genes from wild relatives enables multigene stacking that could enhance durability. Here we use map-based cloning and long-read sequencing to isolate two adjacent nucleotide-binding and leucine-rich repeat (NLR) receptors from wild emmer wheat. Using mutagenesis, gene silencing and genetic transformation, we show that the genes *TdNLR1* and *TdNLR2* oriented head-to-head are both required for *YrTD121*-mediated stripe rust resistance. *TdNLR1* encodes a canonical NLR (CC-NB-ARC-LRR) protein, whereas *TdNLR2* encodes an atypical one (NB-ARC-LRR). Both genes lack an integrated domain previously associated with effector perception, representing an uncommon architecture for paired NLRs in plants. The coiled coil domain of *TdNLR1* triggers cell death and self-associates in planta. *YrTD121* was present in wild emmer but absent in all other *Triticum* species examined. Our work sheds light on the function of paired NLRs in conferring disease resistance and facilitates breeding for resistance.

Nucleotide-binding and leucine-rich repeat (NLR) proteins, a large family of plant immune receptors, directly or indirectly recognize pathogen effectors to activate a defense response[1,2]. Canonical NLRs have a conserved tripartite architecture containing an N-terminal coiled coil (CC) or Toll/interleukin-1 receptor (TIR) domain, a central nucleotide-binding (NB-ARC) domain and a C-terminal leucine-rich repeat (LRR) domain[2]. The CC and TIR domains are associated with effector perception and signal transduction; the NB-ARC domain acts as a molecular switch for NLR activation; and the LRR domain has a role in autoinhibition and defines effector recognition specificity[3–5]. Additional integrated domains in some NLRs act as effector sensors[6].

Single NLRs are sufficient for effector recognition and downstream defense activation. However, certain NLRs function as specialized sensor and helper pairs in which a sensor NLR (sNLR) recognizes a pathogen effector and a helper NLR (hNLR) induces downstream immune responses[7,8]. Most NLR pairs studied to date are genetically linked in

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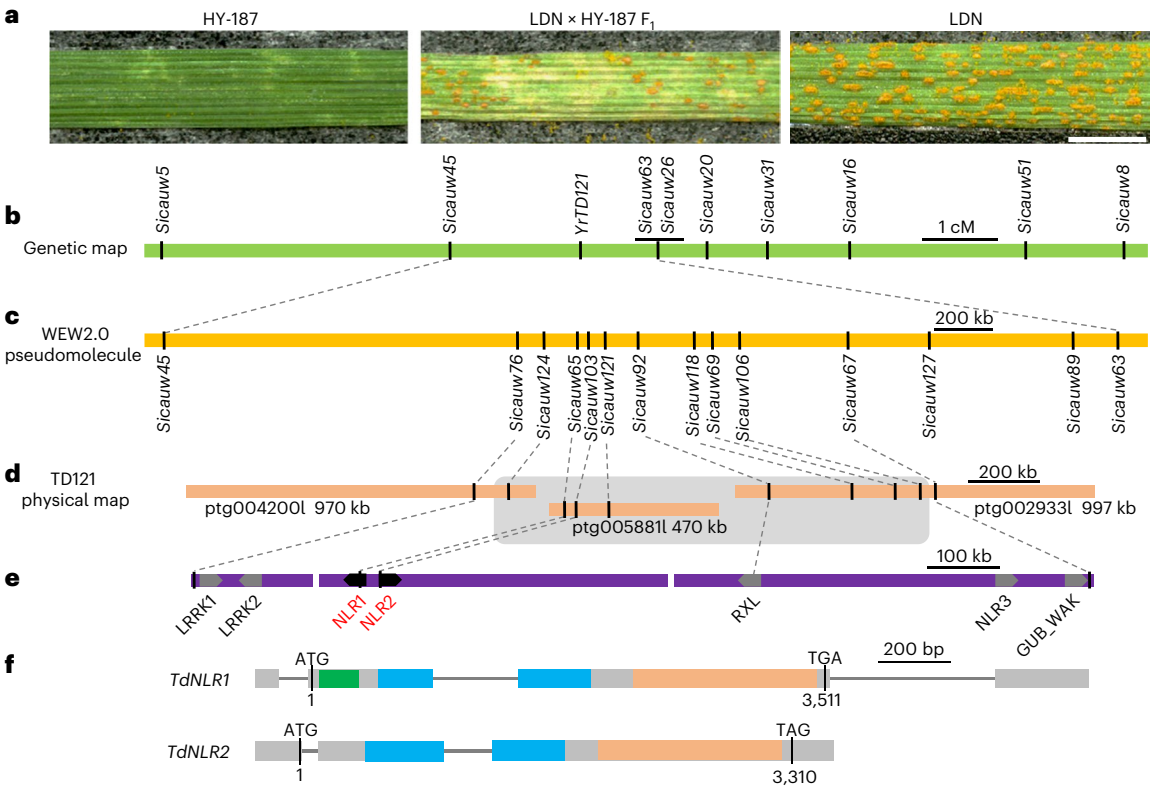

Fig. 1 | Map-based cloning of *YrTD121*. **a**, Leaves of three-leaf stage HY-187, LDN and LDN × HY-187 F_1 plants 14 days after inoculation with *Pst* race CYR34. **b**, Genetic map of *YrTD121* based on F_2 genotypes from cross LDN × HY-187. **c**, Genomic sequence of Zavitan 1BS pseudomolecule corresponding to the *YrTD121* genetic interval. **d**, 1BS HiFi contig assembly in the genetic region of *YrTD121*. The assembly of TD121 HiFi sequences generated three nonoverlapping contigs: ptg004200l (970 kb); ptg005881l (470 kb); and ptg002933l (997 kb). The physical map of the *YrTD121* region is shown in the gray rectangle. **e**, Predicted genes associated with stripe rust resistance in the *YrTD121* region of the TD121 1B pseudomolecule. **f**, Structures of *TdNLR1* and *TdNLR2* in TD121. The rectangular boxes represent exons and the gray lines represent introns. The CC domain is shown in green; the NB-ARC domain is shown in blue; and the LRRs are shown in orange. **a**, Scale bar, 0.5 cm.

head-to-head orientation and function via either negative regulation or receptor cooperation[9–13]. However, some sNLRs function together with multiple hNLRs to form an intricate NLR network[13,14]. sNLRs typically contain integrated domains, such as heavy-metal-associated, kinase, WRKY and BED-type zinc finger domains, typically derived from host proteins targeted by pathogens and used as bait for mediating effector detection[6,15].

Common wheat (*Triticum aestivum* L.) provides approximately 20% of the calories and protein requirements in human nutrition[16]. Stripe rust (or yellow rust) disease caused by *Puccinia striiformis* Westend. f. sp. *tritici* (*Pst*) poses a serious threat to global wheat production. Exploiting resistance (*R*) genes in cultivars is the most efficient and environmentally friendly strategy to control the disease. More than 80 yellow rust resistance genes (*Yr1*–*Yr87*) have been formally documented[17–19]. However, only a few have been cloned, and these encode either singleton NLRs or non-NLR proteins belonging to disparate structural classes[20–22]. The fast-evolving nature of virulent pathogen races means that continuous efforts to find additional *R* genes and deploy them into cultivated wheat are crucial to achieve more durable resistance.

Wild emmer wheat (*Triticum turgidum* ssp. *dicoccoides*), the tetraploid progenitor of common wheat, harbors disease resistance genes that are absent in cultivated wheat[23]. However, isolating *R* genes from this species is hampered by suppressed recombination and structural variations in the genome[24–26]. Recent advances in wheat genomic resources[27,28] and technological breakthroughs in DNA sequencing[29] facilitate the identification of new *R* genes in wheat and its wild relatives[22,24,25]. In this study, we report the cloning of the stripe rust *R* gene *YrTD121* from the wild emmer accession TD121 (TD104059)[30] through positional cloning and PacBio long-read sequence assembly. We used mutagenesis, virus-induced gene silencing (VIGS), CRISPR–Cas9-based genome editing and stable genetic transformation to demonstrate that *YrTD121*-mediated resistance entails the cooperation of a *TdNLR1* and *TdNLR2* gene pair. *YrTD121* was introgressed into cultivated wheat and can be used to improve stripe rust resistance in durum (*T. turgidum* ssp. *durum*) and common wheat.

Results

YrTD121 confers incompletely dominant stripe rust resistance

The wild emmer accession TD121 exhibited high resistance (infection type (IT) = 1–2 on a 0–9 scale) to the *Pst* race CYR34 at the seedling stage; adult plants were immune (IT = 0) to a mixture of *Pst* races in the field (Supplementary Fig. 1a). Tests with gene-based primers revealed the presence of the *R* gene *Yr36* in TD121 (Supplementary Fig. 1b). To exclude the role of *Yr36* in conferring resistance, we screened F_5 progenies from a cross between TD121 and the stripe rust-susceptible durum variety Langdon (LDN), selected the highly resistant line HY-187 lacking *Yr36* and backcrossed it to LDN. While HY-187 was highly resistant to *Pst* race CYR34 (IT = 1–2), F_1 seedlings showed an intermediate response (IT = 4–5; Fig. 1a) and the F_2 population segregated 42 (resistant):66 (intermediate):37 (susceptible) ($\chi^2_{1:2:1} = 0.828$, $P = 0.661$), indicating that resistance was conferred by an incompletely dominant gene, tentatively designated *YrTD121*.

YrTD121 was mapped to a 1.1-Mb interval on chromosome 1BS

To map *YrTD121*, resistant and susceptible RNA bulks, together with the parental pools, were subjected to bulked segregant RNA sequencing

(BSR-seq) analysis[31]. High-quality reads from each bulk and parental pool were retained and mapped to the wild emmer reference genome (Supplementary Table 1). With a filtration criterion of allele frequency difference (AFD) ≥ 0.7, we identified 220 single-nucleotide polymorphisms (SNPs) potentially associated with the resistant phenotype; 130 SNPs were enriched on chromosome (chr) 1B (Supplementary Fig. 2a), with 61 clustered in a 20-megabase (Mb) genomic interval (0–20 Mb) on the short arm (Supplementary Fig. 2b). Six kompetitive allele-specific PCR (KASP) and three PCR markers were developed from SNPs or insertions and deletions (indels) identified within the 0–20 Mb genomic interval and used to genotype 145 F_2 plants derived from LDN × HY-187 (Fig. 1b and Supplementary Table 2). The closest flanking markers, *Sicauw45* and *Sicauw63*, placed *YrTD121* within a 3.5-centimorgan (cM) genetic interval (Fig. 1b), corresponding to a 3.18-Mb physical region in Zavitan reference genome v.2.0 (9.66–12.84 Mb in chr 1B).

Among a segregating F_2 population of 5,256 plants pooled from three crosses, 37 recombinant inbred lines (RILs) (F_{4-6} generations) representing at least nine recombination events were identified between the markers *Sicauw45* and *Sicauw63* (Supplementary Fig. 3 and Supplementary Table 3). Twelve additional markers in this interval anchored the *YrTD121* locus to a physical interval of approximately 1.1 Mb, flanked by *Sicauw76* and *Sicauw67*, in the Zavitan chromosome arm 1BS (Fig. 1c and Supplementary Fig. 3).

PacBio long-read sequencing and candidate gene analysis

We applied PacBio circular consensus sequencing using the HiFi approach, which generated long-read sequences of TD121 DNA and facilitated the cloning of *YrTD121*. Assemblies from four single-molecule real-time (SMRT) cells (approximately 116.17 gigabases (Gb)) generated 20,653 contigs with a total assembly length of 11.09 Gb and N50 contig length of 3.06 Mb (Supplementary Table 4). A local basic local alignment search tool (BLAST) search using markers closely linked to or cosegregating with *YrTD121* as a query identified three nonoverlapping contigs (ptg004200l, 970 kilobases (kb); ptg005881l, 470 kb; and ptg002933l, 997 kb) spanning 1.18 Mb of genomic sequence in the *YrTD121* region (Fig. 1d). Sequence annotation of these contigs revealed seven putative genes encoding protein domains that were previously implicated in plant immunity[1], including two leucine-rich repeat receptor-like protein kinases (*LRRK1* and *LRRK2*), three *NLRs* (*NLR1*, *NLR2* and *NLR3*), an RX-CC_like domain and a wall-associated receptor kinase galacturonan-binding (*GUB_WAK_Bind*) domain (Fig. 1e and Supplementary Table 5).

RNA-seq analysis was performed on the seedlings of the lines HY-45, HY-64 and HY-187 inoculated with *Pst* race CYR34. After mapping the generated reads to the TD121 PacBio contigs, two transcribed *R* genes were identified in the *YrTD121* physical region and designated as *TdNLR1* and *TdNLR2* (Supplementary Fig. 4). These two NLR genes, organized in head-to-head orientation, were prioritized as candidate genes for *YrTD121* (Fig. 1e,f). *TdNLR1* contained two exons and encoded a 961 amino acid residue typical of a CC-NB-LRR protein, whereas *TdNLR2* had three exons and encoded a predicted protein of 1,002 amino acids (aa) with putative NB-ARC and LRR domains. TdNLR1 and TdNLR2 shared ~22% amino acid identity at the protein level. Neither TdNLR1 nor TdNLR2 contained integrated domains previously implicated in effector perception[6].

Functional validation of the candidate genes for *YrTD121*

To investigate the role(s) of *TdNLR1* and *TdNLR2* in resistance to stripe rust, we used ethyl methanesulfonate (EMS) to create susceptible mutants in *YrTD121* in the tetraploid line HY-45 and hexaploid line Mianmai 37 + *YrTD121* (hereafter MM37 + *YrTD121*). The genomic DNA (gDNA) sequences of the two *TdNLR* genes were amplified and sequenced in each susceptible mutant. Nine mutants (four tetraploid, designated M; five hexaploid, designated EMS) carried point mutations that resulted in amino acid substitutions in the TdNLR1 protein (Fig. 2a,b); the mutant lines M94 and EMS143 had the same mutation site in the *TdNLR1* coding sequence (CDS) (C1121T, protein P374L). Eight mutants had point mutations in the *TdNLR2* CDS predicted to cause amino acid changes or premature termination of translation (Fig. 2a,c). These results indicated that mutations in either *TdNLR1* or *TdNLR2* caused loss of resistance to stripe rust in the HY-45 and MM37 + *YrTD121* lines.

To further validate the functions of *TdNLR1* and *TdNLR2* in resistance, we designed barley stripe mosaic virus (BSMV)-induced gene silencing constructs targeting either the NB-ARC or LRR domains of *TdNLR1* and *TdNLR2* (Supplementary Figs. 5 and 6). Inoculation of HY-45 with the silencing constructs led to a 40–85% reduction in transcript levels of the *TdNLR* genes (Fig. 2d,e). *TdNLR1* or *TdNLR2* gene-silenced leaves exhibited reduced stripe rust resistance with the average percentage of leaf areas covered by uredinia ranging from 14.02% to 21.14% (Fig. 2f,g and Supplementary Figs. 5 and 6), whereas the negative control leaves were resistant to stripe rust with no visible uredinia. These results demonstrated that both *TdNLR1* and *TdNLR2* were essential for *YrTD121*-mediated resistance.

Next, we used CRISPR–Cas9-mediated genome editing to knock out *TdNLR1* or *TdNLR2* in HY-45. We designed single-guide RNAs (sgRNAs) to target the CDS of each *TdNLR* gene and generated transgenic lines. Mutation sites in the target region were detected using Sanger sequencing. We identified three mutant lines for *TdNLR1* (*TdNLR1*KO-1 to *TdNLR1*KO-3), carrying 4-bp, 7-bp or 13-bp deletions, and two mutant lines for *TdNLR2* (*TdNLR2*KO-1 and *TdNLR2*KO-2) with a 1-bp insertion and a 6-bp deletion in the target sites, respectively. These mutants were susceptible to *Pst* race CYR34, whereas nonedited HY-45 plants were highly resistant (Fig. 3a), indicating that both *TdNLR1* and *TdNLR2* were required for resistance in HY-45.

We transformed the *Pst*-susceptible hexaploid wheat cultivar Fielder using full-length genomic fragments of *TdNLR1* or *TdNLR2* with native promoters and downstream regulatory sequences (Fig. 3b), as well as the open reading frames of *TdNLR1* or *TdNLR2* driven by the maize (*Zea mays*) ubiquitin promoter (Supplementary Fig. 7a). We also cotransformed the *TdNLR1* and *TdNLR2* constructs with native or ubiquitin promoters into Fielder. T_0 individuals expressing *TdNLR1* or *TdNLR2* individually, and *TdNLR1* and *TdNLR2* together, were identified using PCR; when challenged with *Pst* race CYR34, all T_1 plants expressing both *TdNLR1* and *TdNLR2*, or *TdNLR1* alone, were resistant, whereas plants expressing *TdNLR2* alone were susceptible (Fig. 3c and Supplementary Fig. 7b–d). Three *TdNLR2*-like homologs were identified in Fielder chr 1B (TraesFLD1B01G018900.1), chr 2B (TraesFLD2B01G357400.1) and chr 2D (TraesFLD2D01G344800.1), respectively (Supplementary Fig. 8a). The 1B copy was truncated because of a premature stop codon; the 2B and 2D copies shared low amino acid sequence similarity (approximately 44.0%) with *TdNLR2*, although their predicted protein structures were similar (Supplementary Fig. 8b), indicating a potentially redundant function of *TdNLR2*-like homologs in conferring resistance.

Taken together, the loss of function caused by EMS-mediated, VIGS-mediated and CRISPR–Cas9-mediated mutations and transgenic complementation provided evidence that *TdNLR1* positively contributed to stripe rust resistance and that loss of *TdNLR2* function compromised *TdNLR1*-mediated resistance at the *YRTD121* locus.

Expression analysis

We measured the transcript levels of *TdNLR1* and *TdNLR2* in HY-187 using quantitative PCR with reverse transcription (RT–qPCR). *TdNLR1* and *TdNLR2* were expressed in roots, stems and leaves at the two-leaf stage, with the highest expression levels in leaves (Supplementary Fig. 9a,b). Higher expression levels of both *TdNLR1* and *TdNLR2* were detected in the leaves of *YrTD121* homozygotes than heterozygotes (Supplementary Fig. 9c,d). There was no difference in *TdNLR1* or *TdNLR2* expression levels in mock-inoculated or *Pst*-inoculated plants at 1 day after inoculation and 3 days after inoculation (Supplementary Fig. 9e,f).

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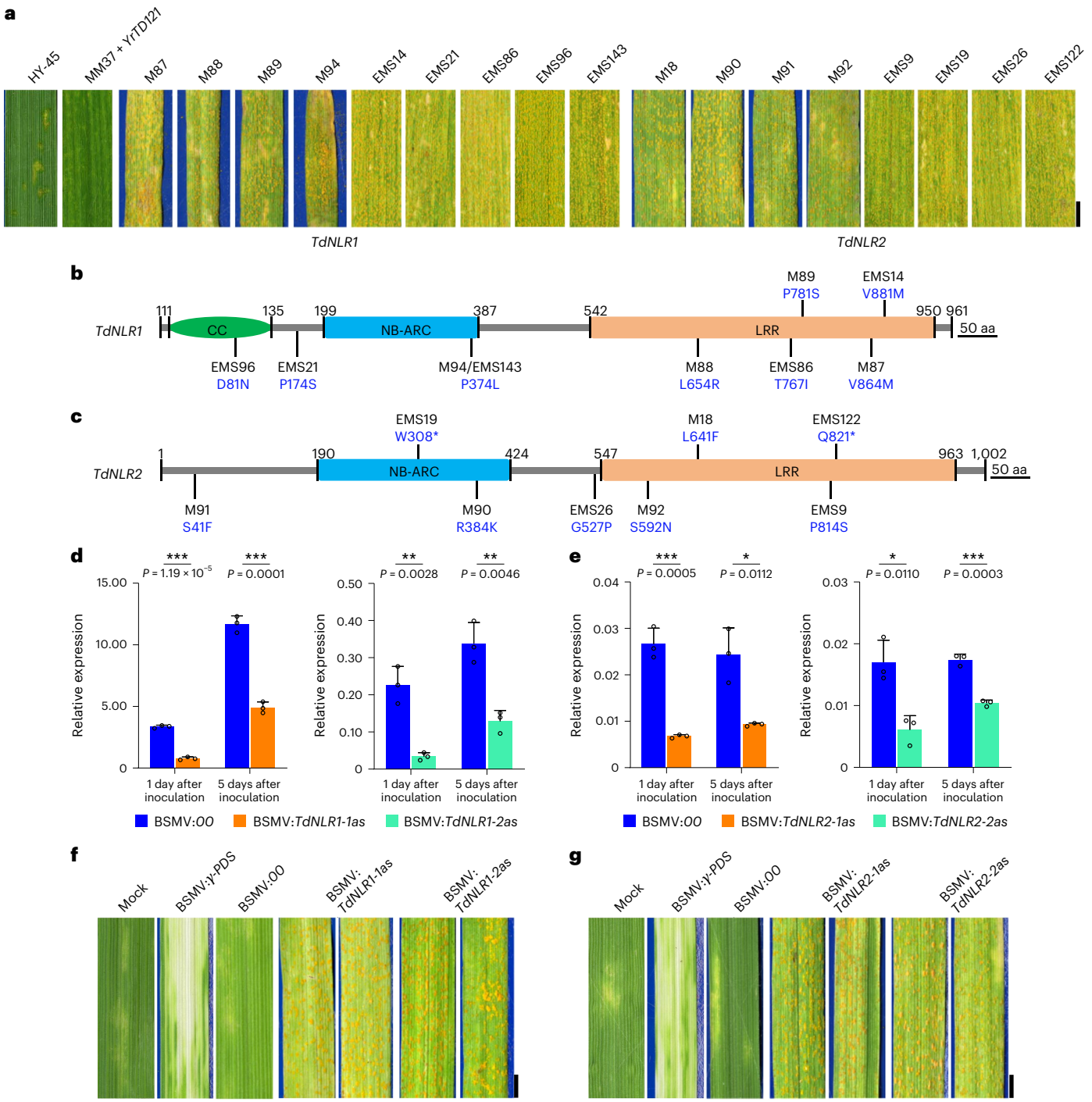

Fig. 2 | Functional validation of *TdNLR1* and *TdNLR2* by mutagenesis and BSMV-induced gene silencing assays. a, Stripe rust responses of tetraploid HY-45 and hexaploid MM37 + *YrTD121* and the respective M and EMS mutants. **b**, Positions of mutations in *TdNLR1* in the M and EMS mutants. **c**, Positions of mutations in *TdNLR2* in the M and EMS mutants. Amino acid changes are shown in blue. *, premature termination of translation. **d**,**e**, Validation of *TdNLR1* and *TdNLR2* using VIGS. Reduced expression levels of *TdNLR1* in leaves inoculated with BSMV:*TdNLR1* (**d**) and BSMV:*TdNLR2* (**e**) are shown. The BSMV:*TdNLR1-1as* and BSMV:*TdNLR1-2as* constructs targeted the NB-ARC domain of TdNLR1. The BSMV:*TdNLR2-1as* and BSMV:*TdNLR2-2as* constructs targeted the NB-ARC and LRR domains of TdNLR2. The error bars represent the mean ± s.d. ($n = 3$ biologically independent samples, each sample with three technical replicates). Statistical significance, as indicated by the *P* value, was determined using a two-tailed Student's *t*-test (*$P < 0.05$; **$P < 0.01$, ***$P < 0.001$). **f**,**g**, Resistance was compromised when either *TdNLR1* (**f**) or *TdNLR2* (**g**) was silenced. Plants treated with BSMV:*γ-PDS* (phytoene desaturase) were used as positive controls to check the efficacy of VIGS. Symptoms of photobleaching were observed in PDS-silenced plants. **a**,**f**,**g**, Scale bar, 0.5 cm.

Haplotype variation and phylogenetic analysis

Screening 906 *Triticum* accessions using gene-specific primers detected the *TdNLR* genes in 17% (62 of 363) of wild emmer accessions from Israel, Syria, Lebanon and Turkey but not in other *Triticum* species or subspecies (Supplementary Table 6). Fifty-seven *T. dicoccoides* accessions possessed both *TdNLR1* and *TdNLR2*. A survey using additional sets of homolog primers identified two accessions carrying *TdNLR1* and *nlr2* and three with *TdNLR2* and *nlr1* (Supplementary Figs. 10 and 11).

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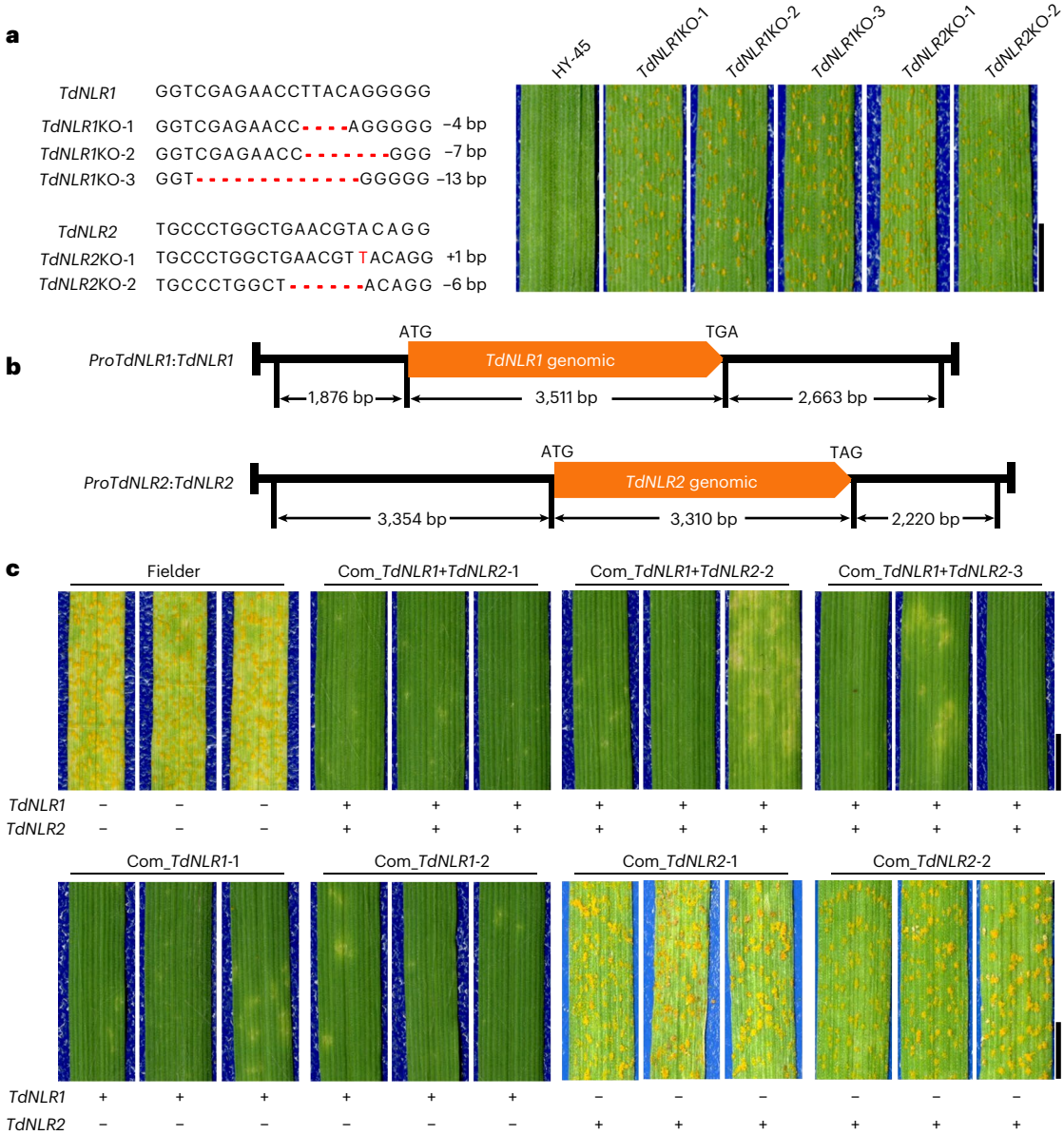

Fig. 3 | Functional validation of *TdNLR1* and *TdNLR2* using CRISPR–Cas9 gene editing and transgenic complementation. **a**, Left, Mutations induced in the *TdNLR1* or *TdNLR2* nucleotide sequences of edited plants. Right, Responses of the *TdNLR1* or *TdNLR2* knockout lines to inoculation with *Pst* race CYR34. **b**, The structures of *ProTdNLR1*:*TdNLR1* and *ProTdNLR2*:*TdNLR2* used for the complementation assays. **c**, Reactions of T_1 transgenic lines expressing *TdNLR1* or *TdNLR2* from the native promoter (Com). PCR results as positive (+) or negative (−) for DNA amplification of the respective *TdNLR* gene are shown below the images. **a**,**c**, Scale bar, 0.5 cm.

Analysis of full-length *TdNLR1* sequences from 59 *T. dicoccoides* accessions identified 16 haplotypes (*TdNLR1_h1* to *TdNLR1_h16*) representing 11 predicted protein variants (Supplementary Fig. 12 and Supplementary Table 7). Proteins from haplotypes *TdNLR1_h4*, *TdNLR1_h5*, *TdNLR1_h15* and *TdNLR1_h16* shared a 21-nucleotide deletion at positions 87–107, causing deletion of seven aa (GVHDDIQ) in the CC domain relative to other haplotypes (Supplementary Figs. 12 and 13). Alignment of 60 *TdNLR2* sequences identified 11 haplotypes (*TdNLR2_h1* to *TdNLR2_h11*) defining ten predicted protein variants (Supplementary Fig. 14 and Supplementary Table 7). *TdNLR2_h8* had a 'TA' insertion in the interdomain region between the NB-ARC and LRR domains; *TdNLR2_h11* had a 649-nucleotide deletion spanning 465 base pairs (bp) of the 5′-UTR and 184 bp of the CDS, starting with ATG (Supplementary Figs. 14 and 15).

The combined TdNLR1 and ten TdNLR2 protein variants represented 20 protein haplotype combinations (PC1–PC20) (Supplementary Table 7). Three accessions shared haplotypes identical to TD121 (PC1). Wild emmer accession PI 487260 (*Yr84*) with haplotype PC2 expressed *TdNLR1* and *TdNLR2*. Thirty-six accessions carrying different PC haplotypes were resistant to stripe rust in the field (Supplementary Table 7). However, some of them also possessed *Yr15* or *Yr36*, which would have been effective. Thirty accessions without *Yr15* and *Yr36*, including haplotypes PC2, PC3, PC4, PC14 and PC19, were resistant to *Pst*, suggesting that they might be functional. PC14 was probably functional according to segregation in an F_2 population derived from a cross of wild emmer WEW110676 (PC14) and LDN (Supplementary Table 8). Accessions carrying PC7, PC8, PC9, PC11, PC13, PC15 or PC20 were susceptible, suggesting that these were nonfunctional haplotypes (Supplementary Table 7).

A BLAST search against the available reference genomes of cereal species using the TdNLR1 and TdNLR2 amino acid sequences as a query returned several hits with different levels of similarity

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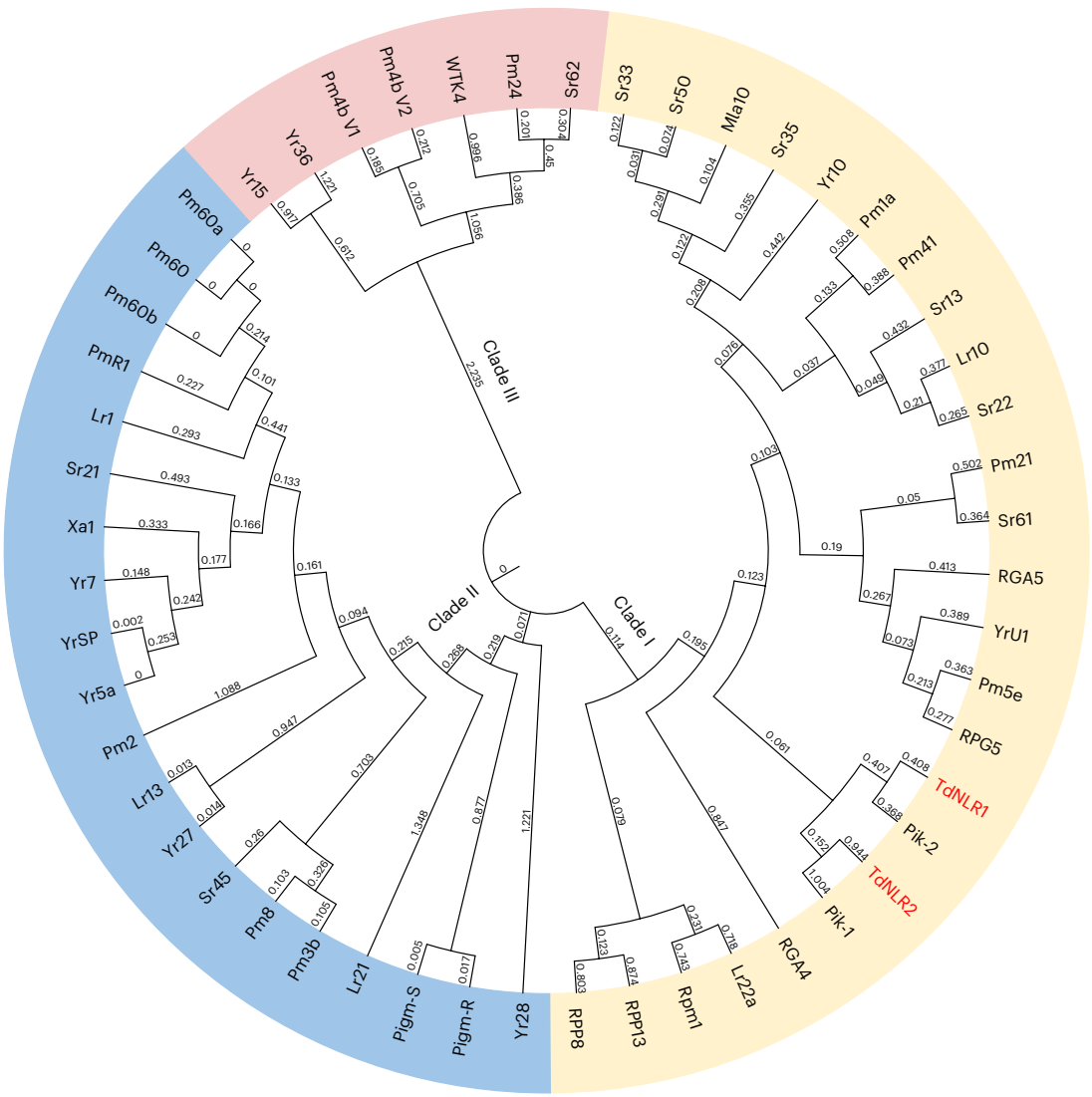

Fig. 4 | Phylogenetic tree based on TdNLR1, TdNLR2 and 50 cloned R proteins in cereal species. Proteins with NLR structure are shaded light yellow (clade I) and light blue (clade II). Nonconventional R proteins without NLR structures are shaded pink (clade III). The numbers on the branches represent branch length.

(Supplementary Tables 9 and 10). Among them, homologs located in homoeologous group 1 chromosomes of tetraploid and hexaploid accessions (1B), as well as the diploids *Secale cereale* (1R), *Aegilops searsii*, *Aegilops bicornis* and *Aegilops longissima* (1S), shared a high amino acid sequence identity (87.0–90.0%) with TdNLR1 (Supplementary Table 9). Other homologs on chromosomes 2B, 2D, 5B, 5R and 5S had less than 75% identity to TdNLR1. By contrast, the best-matched homologs of TdNLR2 were located only in chromosomes 1B, 1R and 1S and shared less than 65% identity with TdNLR2 (Supplementary Table 10). Phylogenetic analysis of TdNLR proteins and putative homologs on group 1 chromosomes in annotated cereal genome assemblies showed that the TdNLR1 protein was most closely related to its ortholog AE.LONG.r1.1SG0000960.1 in *Ae. longissima* (Supplementary Fig. 16); the TdNLR2 protein had a relatively close evolutionary relationship with *Ae. searsii*.TE01.1S01G0017500.1 in *Ae. searsii* (Supplementary Fig. 17), suggesting that the *TdNLR1* and *TdNLR2* genes at the *YRTD121* locus may have evolved from chromosome 1S of the *Aegilops* species.

To determine the phylogenetic relationships of the TdNLR pair with other cloned R proteins in cereal species and wild relatives, we constructed a phylogenetic tree based on the alignment of TdNLR1 and TdNLR2 together with 50 published R proteins (Supplementary Table 11). Three main clades (I–III) were identified, with clades I and II including NLR proteins, and clade III containing other types of R proteins. TdNLR1 and TdNLR2 were in clade I (Fig. 4). The R proteins most closely related to the TdNLR1 and TdNLR2 pair were the rice (*Oryza sativa*) Pik-1 and Pik-2 pair, previously reported to be essential for *Pik*-mediated resistance to rice blast[32]. TdNLR1 shared 43.3% amino acid identity with Pik-2, whereas the similarity between TdNLR2 and Pik-1 was 25.8%, suggesting sequence divergence between these paired NLRs, while both retain critical roles in the regulation of plant immunity.

TdNLR1_CC triggers cell death and self-associates

To investigate the functional and molecular interactions between TdNLR1 and TdNLR2, we transiently expressed full-length TdNLR1 and TdNLR2 proteins and their different domains with a yellow fluorescent protein (YFP) tag in *Nicotiana benthamiana* leaves (Fig. 5a and Supplementary Fig. 18a). Expression of the TdNLR1_CC domain with YFP fused to the C terminus induced cell death, which became visible at 72 h after infiltration (Fig. 5b,c); however, no cell death was observed when the YFP was fused to the N terminus of the CC domain (Supplementary Fig. 19). Intriguingly, compared to TdNLR1_CC, expression of the CC domain of TdNLR1_h4 (TdNLR1_CC^{h4}) or an artificial chimeric

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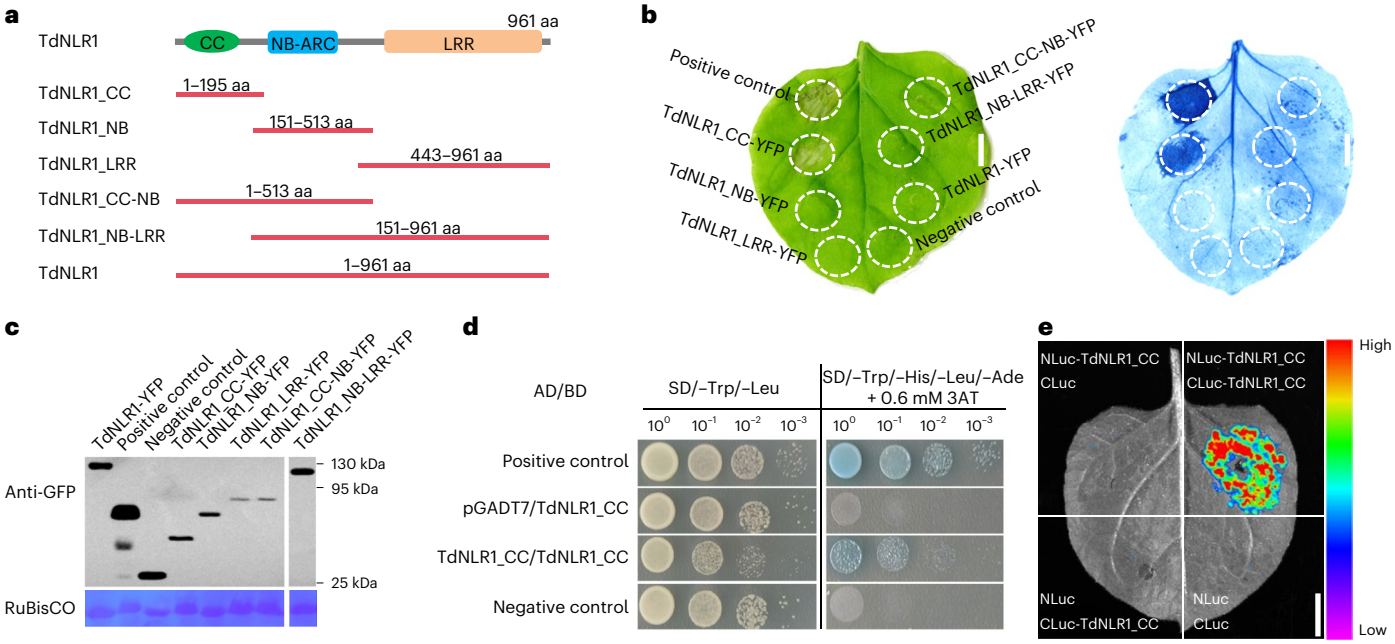

Fig. 5 | Cell death assay and self-association of the TdNLR1 CC domain. **a**, Schematic showing the different domains of TdNLR1 (CC, NB, LRR, CC-NB, NB-LRR) and the full-length protein used in the cell death assays in *N. benthamiana* leaves. **b**, Transient expression of different TdNLR1 domains in *N. benthamiana*. Infiltrated leaves (left) were stained with trypan blue (right) to observe cell death 72 h after infiltration. YFP was fused to the C terminus of different domains. The positive control was the wheat TaZFP5B protein. The negative control was YFP. **c**, Immunoblotting was used to detect the expression of constructs in *N. benthamiana* leaves. Coomassie brilliant blue staining of RuBisCO is shown below as the loading control. **d**,**e**, Self-association of the CC domain of TdNLR1 in the yeast two-hybrid (**d**) and luciferase complementation imaging (**e**) assays. The sequence of TdNLR1_CC for the yeast two-hybrid assays was fused to the Gal4 transactivation domain (AD) or Gal4 DNA-binding domain (BD). Recombinant constructs were cotransformed into Y2HGold, added to synthetic dropout (SD) medium lacking tryptophan and leucine (SD/−Trp/−Leu) and on SD medium lacking tryptophan, histidine, leucine and adenine (SD/−Trp/−His/−Leu/−Ade). The SD/−Trp/−His/−Leu/−Ade medium was supplemented with 0.60 mM 3-aminotriazole (3AT) and X-α-gal. The positive control was pGADT7-T/pGBKT7-53; the negative control was pGADT7-T/pGBKT7-lam. Images were captured 48 h after infiltration. All experiments were repeated three times with consistent results. CLuc, *Cypridina* luciferase; NLuc, NanoLuc. **b**,**e**, Scale bar, 1 cm.

CC variant TdNLR1_CC^{del} with a seven-amino-acid (GVHDDIQ) deletion impaired the ability of CC to trigger cell death in *N. benthamiana* leaves (Supplementary Fig. 20). We observed no cell death after expression of full-length TdNLR1 or the NB, LRR, CC-NB or NB-LRR domains of TdNLR1 (Fig. 5b and Supplementary Fig. 19). Expression of full-length TdNLR2 or different domains of TdNLR2 failed to induce a response in *N. benthamiana* leaves, suggesting that they had no cell-death-inducing activity (Supplementary Fig. 18b–d). Yeast two-hybrid and split luciferase assays in *N. benthamiana* to test the interaction between TdNLR1 and TdNLR2 also failed to detect direct physical interaction of full-length TdNLR1 and TdNLR2 (Supplementary Fig. 21).

Self-association of the CC domain of NLR proteins has a role in triggering downstream immune signals[5,9,33]. Thus, we examined whether the CC domain of TdNLR1 self-associated using yeast two-hybrid and luciferase complementation imaging experiments. The CC domain of TdNLR1 indeed self-associated in vitro and in planta (Fig. 5d,e). However, we observed no direct interaction of TdNLR1 with itself or with its CC, NB, LRR, CC-NB or NB-LRR domains using yeast two-hybrid assays (Supplementary Fig. 22).

Introgression of *YrTD121* into hexaploid wheat

HY-187 was backcrossed to the common wheat cultivar Chuanmai 42 (CM42) and *YrTD121* was selected using the flanking markers *Sicauw45* and *Sicauw26* (Supplementary Fig. 23). The selected homozygous BC_3F_5 line CM42 + *YrTD121* showed higher resistance (IT = 1–2) to stripe rust pathogen than shown by CM42 (IT = 5–6; Fig. 6). There was no significant difference between CM42 + *YrTD121* and CM42 in agronomic traits (Fig. 6). To accelerate future deployment and pyramiding of *YrTD121* with other *R* genes, we developed the marker *KASP-FM* (Supplementary Fig. 24 and Supplementary Table 12) to facilitate marker-assisted selection.

Discussion

Wild emmer wheat is an important resource for improving tolerance of cultivated wheat to biotic and abiotic stresses[23]. To date, only a few wild emmer *R* genes have been fully characterized, such as the wheat tandem kinase 1 (*WTK1*) gene *Yr15* (ref. 34), the kinase-START 1 (*WKS1*) gene *Yr36* (ref. 35), the tandem kinase 7 (*WTK7-TM*) gene *Pm36* (ref. 25) and the NLR genes *Pm26* (ref. 36), *Pm41* (ref. 26), *TdPm60* (ref. 37) and *Pm69* (ref. 24). In this study, we cloned the stripe rust resistance gene *YrTD121* and confirmed that two genetically linked *TdNLR* genes are both required for *YrTD121*-mediated resistance. The *TdNLR* pair at the *YRTD121* locus provides an additional source of resistance.

The *YrTD121* location in chromosome 1BS differs from the location of most previously reported *Yr* genes (Supplementary Fig. 25), except for *Yr84* (ref. 18) identified in wild emmer accession PI 487260. Our survey of more than 300 *T. dicoccoides* accessions identified *TdNLR1* and *TdNLR2* in PI 487260. The NLR pair *CNL* (*TdNLR1*) and *NL* (*TdNLR2*) in PI 487260 conferring stripe rust resistance at the *Yr84* locus is reported in the companion paper[38]. Sequence comparisons between *YrTD121* and *Yr84* revealed three SNPs in *TdNLR1* (A933C, T2184C and A3221G) and one SNP in *TdNLR2* (A441G). The SNPs T2184C and A3221G are predicted to result in aa changes (L520S and T866A) in TdNLR1; SNP A441G causes a K110E substitution in TdNLR2 (ref. 38), implying that *YrTD121* and *Yr84* are functional haplotypes. The absence of *YrTD121* in the gene pool of cultivated tetraploid and hexaploid wheat suggests that it was either left behind or lost during wheat domestication. Among wild emmer populations, we identified 20 TdNLR protein haplotype combinations

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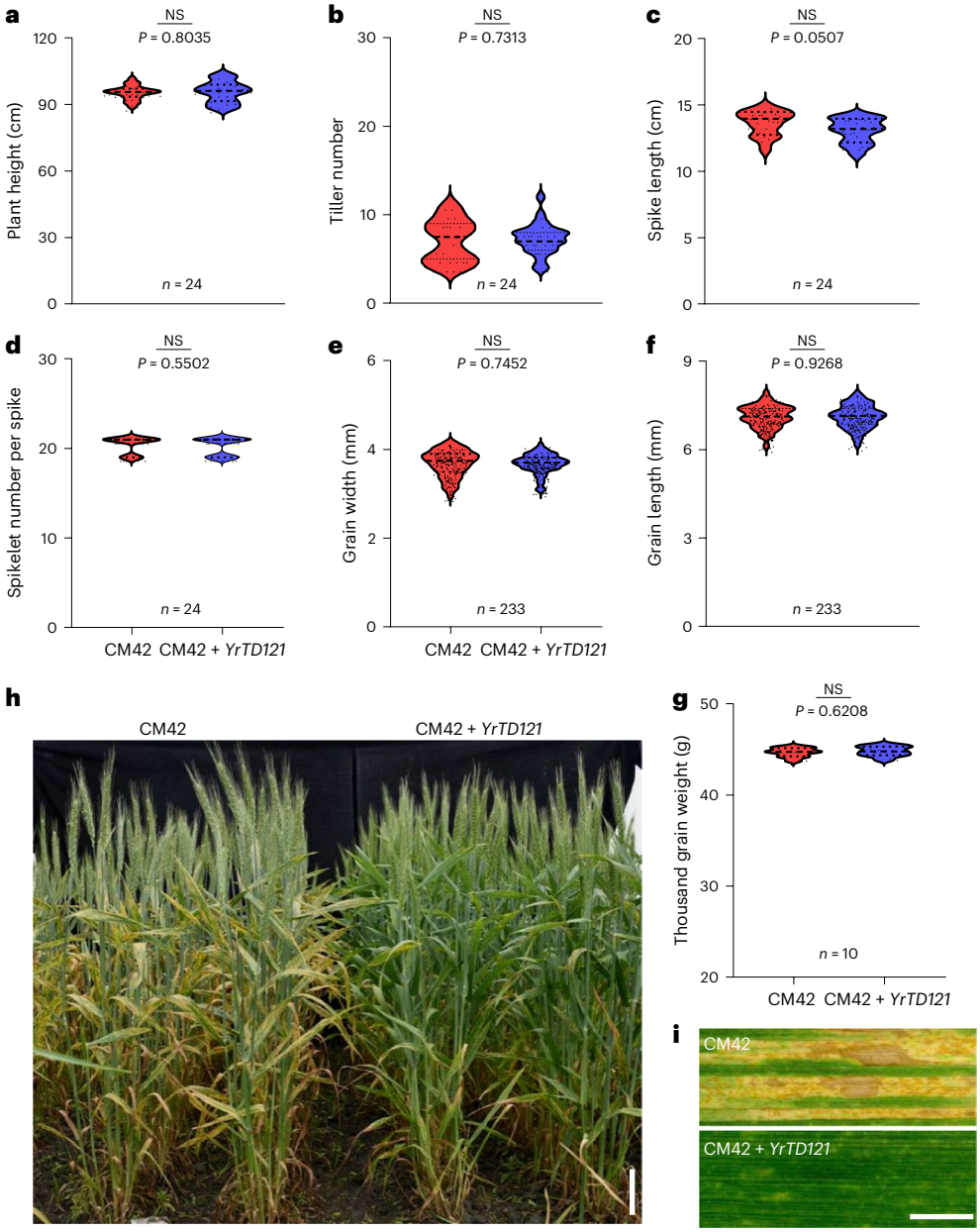

Fig. 6 | Agronomic traits of CM42 and CM42 + *YrTD121*. **a**–**g**, Wheat plants grown under disease-controlled field conditions. Comparisons of plant height (**a**), tiller number (**b**), spike length (**c**), spikelet number per spike (**d**), grain width (**e**), grain length (**f**) and thousand grain weight (**g**). The shape of each violin plot shows the distribution of the trait values. The black dots represent single data points. Statistical significance, as indicated by the *P* value, was assessed using a two-tailed Student's *t*-test. NS, not significant ($P > 0.05$). **h**,**i**, The stripe rust responses of CM42 and CM42 + *YrTD121* in the field. **h**, Scale bar, 10 cm. **i**, Scale bar, 1 cm.

for the *YRTD121* locus; the resistant haplotypes PC1 (*YrTD121*) and PC2 (*Yr84*) accounted for 8% of accessions. The susceptible haplotype PC7 (*TdNLR1_h4* + *TdNLR2_h6*) accounted for up to 30.6% of accessions. The seven aa (GVHDDIQ) deletion in the CC domain of TdNLR1_h4 impaired the cell death-inducing activity probably causing a susceptible host response to stripe rust (Supplementary Fig. 20 and Supplementary Table 7). Other haplotypes, such as PC3, PC4, PC14 and PC19, found in wild emmer require further genetic and functional analysis.

The direct progenitor of the wheat B subgenome diverged from a close relative of *Aegilops speltoides* 4.49 Ma[39]. The closest homologs of *TdNLR1* and *TdNLR2* were identified in the *Sitopsis* species group, such as *Ae. longissima* and *Ae. searsii*, suggesting an ancient origin of a single *NLR*. However, *TdNLR1* and *TdNLR2* at the *YRTD121* locus are highly diverged, suggesting that the *TdNLR* pair was unlikely the result of gene duplication. We found two wild emmer accessions harboring *TdNLR1* and *nlr2* allelic sequences and three accessions carrying *TdNLR2* and *nlr1* allelic sequences. Notably, the *TdNLR1* and *TdNLR2* combination was more prevalent than these genotypes in wild emmer populations. Therefore, a 'true love' model[40] could apply to the origin of the functional *TdNLR1* and *TdNLR2* pair at the *YRTD121* locus in wild emmer, where initially they came together through random evolutionary events that provided coregulatory benefits, which enabled them to resist certain pathogens.

In most cases, a singleton NLR perceives an effector and subsequently forms a resistosome to initiate cell death[41,42]. Recent data suggest that some NLRs function in genetically linked 'sensor–helper' pairs or in genetically unlinked sensor–helper networks[6,9,10]. In rice and *Arabidopsis*, for instance, the NLR pairs *Pik1* and *Pik2* (ref. 32), *RGA4* and

RGA5 (ref. 9), *CHS3* and *CSA1* (ref. 43) and *RRS1* and *RPS4* (ref. 12) function together. Wheat *Lr10* requires the genetically linked RGA2 to mediate leaf rust resistance[10]. A genetically linked atypical NLR pair *TdCNL1* and *TdCNL5* in wild emmer confers powdery mildew resistance at the *PM26*/*MLIW170* locus[36]. We showed that the genetically linked *TdNLR1* and *TdNLR2* pair, in a head-to-head orientation, in wild emmer confers stripe rust resistance. Random introduction of *TdNLR1* and *TdNLR2* into Fielder conferred resistance and indicated that the head-to-head orientation of the gene pair might not be critical for resistance function, whereas the genetic linkage mechanism of *TdNLR1* and *TdNLR2* may facilitate adaptive coevolution and genetic cosegregation of the gene pair to cope with pathogen infection[44]. Transgenic Fielder plants expressing *TdNLR1* alone or both *TdNLR1* and *TdNLR2* were resistant to stripe rust, whereas *TdNLR2* alone did not confer resistance, suggesting that *TdNLR2* may act as an hNLR functioning coordinately with *TdNLR1*. Previous reports showed that the TIR-NB-LRR immune receptor SOC3 partnered with the TIR-NB proteins CHS1 or TN2 to monitor the homeostasis of the E3 ligase SAUL1 (ref. 45); the sNLRs *Rx* and *Bs2* can signal interchangeably via multiple paralogs of hNLRs, including *NRC2*, *NRC3* or *NRC4* (ref. 7) to activate immune responses. Given that expressing *TdNLR1* alone conferred resistance, we hypothesize that the paralogous copies of *TdNLR2* from chromosomes 2B or 2D in Fielder may contribute redundantly to *TdNLR1*-mediated immunity. Further studies are required to decipher the regulatory network of *TdNLR1*-mediated resistance.

Most studied NLR pairs possess a typical NLR and an atypical NLR with an integrated domain and function as heterologous pairs in plant immunity[6,9,11,12]. In our study, TdNLR1 had a typical NLR structure, whereas TdNLR2 was an N-terminally truncated NLR without an integrated domain. Our findings, together with those in the companion paper[38], demonstrate a functional NLR pair without an integrated domain in the wheat stripe rust pathosystem. Although both TdNLR1 and TdNLR2 are required to confer *YrTD121*-mediated stripe rust resistance, we failed to detect a physical interaction between them. In the companion study, a ligand-independent interaction through the LRR domains of the TdNLR1 and TdNLR2 pair was detected[38], a feature that is distinct from the NLR pairs that form heterocomplexes and interact through their CC or TIR domains[9,43]. Future isolation of the *AvrYrTD121* effector and an investigation of their interactions will contribute to our understanding of the mechanism of *YrTD121* in pathogen recognition and immunity.

A preliminary evaluation of the *YrTD121* introgression line CM42 + *YrTD121* in the field experiment showed no significant adverse effects on agronomic and yield traits in the absence of disease, implying the potential value of the gene in wheat breeding. Although *YrTD121* conferred resistance to multiple *Pst* races[18,30,38], it is possible that its resistance will be overcome by currently rare or new virulent *Pst* races. Pyramiding of *R* genes is a promising strategy to maintain adequate and sustainable resistance to plant disease[46,47]. Therefore, *YrTD121* should be pyramided with slow-rusting resistance genes, such as *Yr18* (ref. 48), *Yr36* (ref. 35), and *Yr46* (ref. 49), with different modes of action to achieve more durable resistance. The generated introgression lines and molecular markers developed in this study will facilitate the selection of *YrTD121* in breeding programs.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-025-02207-0.

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Methods

Segregating populations and assays of stripe rust response

Wild emmer accession TD121 (TD104059), originally collected from Avne-Eitan, Israel, was provided by The Lady Barbara Davis Wild Cereal Gene Bank, at the Institute of Evolution, University of Haifa, Israel. TD121 is highly resistant to Israeli *Pst* isolate no. 5006 (race 38E134)[30] and a mixture of prevalent Chinese *Pst* races (CYR32, CYR33, CYR34, Zhong 4 and HY46-8). The tetraploid wheat LDN is highly susceptible to *Pst*. Initially, TD121 was crossed with LDN. The F_5 lines HY-45, HY-64, HY-80 and HY-187 homozygous for *YrTD121* (selected by the markers *WKS1_150F* and *WKS1_620R* to ensure the absence of *Yr36*) were backcrossed with LDN to generate the segregating populations A69, A70, A71 and A68, respectively. The A68 population (LDN × HY-187) of 145 plants was used for the preliminary mapping of *YrTD121*. The remaining three populations totaling 5,256 F_2 plants were used to construct a high-resolution genetic map of *YrTD121*.

Stripe rust tests were carried out in a growth chamber or in experimental fields at Sichuan Agricultural University. The highly virulent Chinese *Pst* race CYR34 was used to assess the seedling response of *YrTD121*-containing lines compared with susceptible LDN. Plants at the 2–4 leaf stage were inoculated using a previously described protocol[46]. Disease responses were evaluated 14–18 days after inoculation using a 0–9 IT scale[50]. Briefly, plants with IT 0–3 were considered resistant; 4–6, intermediate; and 7–9, susceptible.

Evaluations of the parental lines, F_1 and F_2, and further progenies of segregating populations in the field were performed at Sichuan Agricultural University. Individual plants were spaced 10 cm apart within 1.5-m rows, with 30 cm between rows. Avocet Susceptible or Mingxian 169 were used as susceptible checks planted perpendicular and adjacent to the rows of the lines tested. Plants at the tillering stages were inoculated with a mixture of prevalent Chinese *Pst* races (CYR32, CYR33, CYR34, Zhong 4 and HY46-8) provided by the Research Institute of Plant Protection, Gansu Academy of Agricultural Sciences, Lanzhou. Disease responses were recorded three times at 7-day intervals when the susceptible checks showed full disease development on the flag leaves. Final ratings were used for the analysis.

BSR-seq analysis and genetic map construction

The LDN × TD121 F_2 population was evaluated for stripe rust response in the field. Leaf tissues from 27 highly resistant (IT = 0–1) and 27 highly susceptible (IT = 8–9) F_2 plants testing negative for *Yr36* were separately pooled for RNA preparation. Total RNA samples extracted from the parental lines; resistant and susceptible bulks were subjected to RNA-seq on an Illumina HiSeq 4000 platform at Beijing Novogene Technology (www.novogene.com/). Raw data were trimmed to remove low-quality reads and adapter sequences were added using Trimmomatic (v.0.36)[51]. High-quality clean reads were aligned to the wild emmer reference genome sequence v.2.0 (ref. 52) using STAR (v.2.5.1b)[53]. Unique, high-confidence alignments were used to call SNPs and indel variants using GATK (v.3.6)[54]. The allele frequency of each variant and AFD[55] between resistant and susceptible bulks were calculated to identify SNP variants. AFD was defined as the absolute value after subtracting the allele frequency of the resistant bulk from that of the susceptible bulk. Variants with a threshold of AFD ≥ 0.7 were considered associated with resistance and selected for marker development. The resistance-associated SNPs were used to develop KASP markers, which were tested for polymorphisms between the parental lines and the resistant and susceptible DNA bulks. In addition, new indel markers were developed using sequences from the tetraploid wheat reference genome to construct a high-resolution genetic linkage map.

Initial screening of the A68, A69, A70 and A71 populations for recombination events in the *YrTD121* region was performed using the flanking markers *Sicauw5* and *Sicauw8*. Homozygous F_{4-6} RILs were selected and used to fine-map *YrTD121*. These RILs were evaluated in the field for response to stripe rust.

DNA extraction, HiFi library preparation and sequencing

SMRTbell libraries were constructed for sequencing according to the PacBio standard protocol (PacBio). Briefly, high-molecular-weight gDNA for PacBio circular consensus sequencing (CCS) was extracted from the leaves of TD121 seedlings using the cetyltrimethylammonium bromide method[56], followed by purification with a QIAGEN Genomic Kit. DNA purity was assessed using a NanoDrop One UV-Vis spectrophotometer (Thermo Fisher Scientific), and DNA concentration was measured with a Qubit 4 Fluorometer (Invitrogen). Fifteen micrograms of DNA was sheared to an expected target size (~15 kb) using a g-TUBE (Covaris), followed by purification with AMPure PB beads (PacBio). Target DNA fragments were screened using a BluePippin electrophoresis system (Sage Science); the size of library fragments was checked using an Agilent 2100 instrument (Agilent Technologies). Four SMRT cells were sequenced using the CCS mode in a PacBio Sequel II instrument with a 30-h movie by Haorui Genomics.

Sequence assembly and physical map construction

Raw data were processed using CCS v.6.4.0 to produce a highly accurate consensus sequence (HiFi reads). Bedtools v.2.25 (https://bedtools.readthedocs.io/) was used to extract FASTQ records from CCS BAM files. HiFi reads from four CCS cell reads were merged and assembled using hifiasm v.0.16.1-r375 (https://github.com/chhylp123/hifiasm) with default parameters. A local BLAST search was performed using markers closely linked to, or cosegregating with, *YrTD121* as query to identify contigs spanning the *YrTD121* physical region.

Candidate gene identification

Putative genes were predicted with FGENESH (www.softberry.com/berry.phtml) and confirmed using the BLAST analysis tools available at the NCBI (www.ncbi.nlm.nih.gov) and EnsemblPlants (http://plants.ensembl.org). The online CD-Search interface (www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi) and LRR domain prediction tool (https://lrrpredictor.biochim.ro/) were used to search for potential domains associated with disease resistance. Transcriptome analyses to identify expressed genes were performed on *Pst* CYR34-infected leaf tissue from HY-45, HY-64, HY-80 and HY-187 plants at 1 day after inoculation. The generated clean reads were initially aligned to the reference genome of Zavitan (RefSeq v.2.0) using Bowtie 2 v.2.2.5 (ref. 57). The unique mapped read counts were used in fragments per kilobase per million mapped read calculations to represent the expression level. When the TD121 PacBio sequence assembly was available, clean RNA-seq reads were further aligned to the 1.18-Mb TD121 contig sequence of the *YrTD121* region using HISAT2 v.2.2.1 (https://daehwankimlab.github.io/hisat2/) with default parameters. The alignment file in BAM format was indexed using Samtools (www.htslib.org/) and displayed using the Integrative Genomics Viewer[58]. Potential open reading frames in the candidate transcripts were defined based on the RNA-seq data and verified using PCR amplification from the complementary DNA of TD121 using gene-specific primers (Supplementary Table 2).

EMS mutagenesis

EMS mutagenesis of tetraploid HY-45 and hexaploid MM37 + *YrTD121* was performed using a concentration of 0.5% EMS (Coolaber). Briefly, more than 4,000 seeds from each line were soaked in water for 8 h and then incubated for 16 h with shaking (100 rpm) at room temperature in the EMS solution. Before sowing, the seeds were washed twice with 10% sodium thiosulfate and rinsed with running water at room temperature for 4 h.

EMS-treated M_1 plants were grown in an experimental field at Sichuan Agricultural University and screened for stripe rust resistance using the urediniospores of *Pst* races CYR32, CYR33, CYR34, Zhong4 and HY46-8. Taking advantage of the incompletely dominant inheritance of *YrTD121*, spikes were collected from M_1 plants with reduced stripe rust resistance in a field test, assuming they were heterozygous

for *YrTD121* mutations. Then, 10–15 plants of each M_2 family were inoculated with a mixture of *Pst* races CYR32, CYR33, CYR34, Zhong 4 and HY46-8 in the field. The M_3 seeds collected from susceptible M_2 plants were reevaluated with race CYR34 in a growth chamber. Finally, all susceptible mutants identified were confirmed using the *YrTD121*-flanking markers *Sicauw26* and *Sicauw76* to rule out the possibility of seed contamination.

BSMV-induced gene silencing

To silence candidate *TdNLR* genes in HY-45, gene-specific fragments were amplified and inserted into the BSMV-VOX vector. Different vector combinations were linearized and transcribed into RNA in vitro. The transcripts of each vector (α, β, γ or recombinant γ-gene) were mixed in a 1:1:1 ratio and then added to the FES buffer. The mixture was inoculated on the second leaf of seedlings at the third-leaf stage; plants inoculated with BSMV:*γ-PDS* were used as positive controls, while plants treated with BSMV:*00* and FES buffer (mock) were used as the negative controls. Plants infected with virus were incubated at 28 °C for 1 day, then grown under a 16 h at 28 °C in light–8 h at 25 °C in darkness regime. After 10 days, leaves with mild chlorotic mosaic symptoms were inoculated with *Pst* race CYR34 and maintained in a growth chamber (16 h at 15 °C in light–8 h at 10 °C in darkness). The inoculated leaves were sampled at 1 and 5 days after inoculation to assess the efficiency of silencing. Stripe rust symptoms were recorded and photographed 16 days after inoculation. The digital images (each approximately 4-cm leaf segment) were further analyzed using color thresholds in ImageJ (https://imagej.net/ij/) as described previously[59] to quantify the percentage of leaf surface covered with sporulating uredinia.

Wheat transformation

The CRISPR–SpCas9 system was used to validate the function of the proteins encoded by the *TdNLR* genes. Two sgRNAs, sgRNA1 (PAM guide sequence 5′-ACGTCAGCGACCGGCGCACCAGG-3′) and sgRNA2 (PAM-guide sequence 5′-TACGGGGTCGAGAACCTTACAG-3′), were developed to target the exon 1 sequence of *TdNLR1*; sgRNA3 (PAM-guide sequence 5′-ATGCCCTGGCTGAACGTACAGGG-3′) and sgRNA4 (PAM-guide sequence 5′-AGATACAGGCATCTTCTGCTGGG-3′) were developed to target the exon 2 sequence of *TdNLR2*. These sgRNAs were cloned into the pWMBX110-SpCas9 vector[60]. The resulting constructs were introduced into HY-45 by *Agrobacterium tumefaciens*-mediated transformation[61]. Edited M_1 plants were selected using PCR amplification, Sanger sequenced and challenged with *Pst* race CYR34.

The CDS of *TdNLR1* (2,886 bp) and *TdNLR2* (3,009 bp) were amplified and inserted downstream of the maize (*Z. mays*) ubiquitin (*Ubi*) promoter in a *pLGY-OE3* vector to generate the overexpression vectors *ProUbi:TdNLR1* and *ProUbi:TdNLR2*. A genomic fragment of *TdNLR1* containing 3,511 bp of the coding and intron regions, 1,876 bp upstream of the start codon, and 2,663 bp downstream of the stop codon, and a genomic fragment of *TdNLR2* consisting of 3,354 bp upstream, 3,310 bp of the coding and intron regions, and 2,220 bp of the downstream sequence were cloned into the binary vector *pCAMBIA-1300* to develop the complementary constructs *ProTdNLR1*:*TdNLR1* and *ProTdNLR2*:*TdNLR2*. The constructs were sequenced to confirm the correct DNA fragment insertions. The constructs *ProUbi*:*TdNLR1* and *ProUbi*:*TdNLR2* or *ProTdNLR1*:*TdNLR1* and *ProTdNLR2*:*TdNLR2* were introduced separately or cotransformed into the susceptible wheat cultivar Fielder using the PureWheat technique developed by the Japan Tobacco Company. Positive T_0 transgenic plants were identified using PCR with the specific primers *TdNLR1-4* and *TdNLR2-4* (Supplementary Table 2); T_1 plants for each transgenic event were evaluated for stripe rust response.

RT–qPCR analysis

Total RNA was isolated from plant tissues (leaves, roots and stems) using the TRIzol reagent (TIANGEN). First-strand complementary DNA was synthesized using a Takara PrimeScript RT Reagent Kit and qPCR was conducted on a Bio-Rad CFX96 Touch Real-Time PCR System (Bio-Rad Laboratories) using the SYBR Green Master Mix (Takara Bio). Transcript levels were quantified using the $2^{-\Delta C_T}$ method[62]. *TaGAPDH* was used as the internal reference. Three biological replicates were used for each data point. Expression data were further analyzed using Prism v.9.0 (GraphPad Software).

Haplotype analysis

The gene-specific markers *TdNLR1-2* and *TdNLR2-1* (Supplementary Table 2) were used to examine the distribution of *TdNLR1* and *TdNLR2* among various Triticeae species. The entire regions of more than 3.5 kb for *TdNLR1* and 3.6 kb for *TdNLR2* were obtained from 62 *T. dicoccoides* accessions using PCR amplification or whole-genome resequencing. In addition, *nlr* homologous sequences in the accessions carrying either *TdNLR1* or *TdNLR2* alone were obtained using PCR amplification. PCR was performed using a Veriti 96-Well Fast Thermal Cycler (Applied Biosystems) in 20-µl reaction volumes containing 10 µl 2× Phanta Super-Fidelity DNA Polymerase (Vazyme), 100 ng of gDNA and 10 µM of each primer. DNA amplification was performed as follows: 3 min at 95 °C; 35 cycles of 15 s at 95 °C; 55–60 °C for 15 s (depending on the primer annealing temperature); 72 °C for 30 s per kb; and a final extension of 5 min at 72 °C. The amplified PCR products were sequenced using the Sanger chain termination method. Sequence assembly was conducted using the Lasergene package (DNASTAR; www.dnastar.com/). Sequences were analyzed using DNAMAN 8 (Lynnon Biosoft) and aligned using BioEdit[63]. *T. dicoccoides* accessions harboring different *TdNLR* haplotypes were scored for response to stripe rust in the field.

Phylogenetic analyses

ClustalW in MEGA 11 (www.megasoftware.net/) was used for multiple sequence alignments with default settings. A phylogenetic tree was constructed using the maximum likelihood method with 500 bootstrap replicates. Trees were exported as Newick formatted flat files and visualized using iTOL (https://itol.embl.de/). The sequences of the cloned R proteins were downloaded from the NCBI database. TdNLR homologs in cereals (Supplementary Tables 9 and 10) were identified by aligning the TdNLR1 and TdNLR2 protein sequences to the cereal whole-genome assemblies using the WheatOmics1.0 (http://202.194.139.32/jbrowse.html) platform[64].

Cell death assays in *N. benthamiana*

A. tumefaciens strain GV3101 (pSoup-p19) carrying the target constructs was infiltrated into *N. benthamiana* leaves. Infiltrated leaves were photographed 72 h after infiltration and then stained with trypan blue solution to visualize cell death. Total proteins were extracted from the infiltrated *N. benthamiana* leaves before cell death and were detected using immunoblotting and anti-green fluorescent protein (GFP) (1:5,000 dilution, cat. no. D110008, Sangon Biotech) and horseradish peroxidase-conjugated goat anti-mouse IgG (1:10,000 dilution, cat. no. D110087, Sangon Biotech).

Yeast two-hybrid assays

The respective fragments of *TdNLR1* and *TdNLR2* were fused with vectors pGBKT7 and pGADT7 of the yeast two-hybrid system[65]. The recombinant constructs were cotransformed into yeast strain Y2HGold and spotted onto SD medium SD/−Trp/−Leu, SD/−Trp/−His/−Leu and SD/−Trp/−His/−Leu/−Ade to examine protein–protein interactions. Records and photographs were made at 48 h after incubation.

Luciferase complementation imaging assays

Fragments encoding the TdNLR1 and TdNLR2 proteins were fused with *pCAMBIA-35S-nLuc* and *pCAMBIA-35S-cLuc* to generate *cLuc-TdNLR1*, *nLuc-TdNLR1*, *cLuc-TdNLR2* and *nLuc-TdNLR2* constructs to determine protein interactions. A fragment encoding the CC domain of TdNLR1

was fused with *pCAMBIA-35S-nLuc* and *pCAMBIA-35S-cLuc* to generate the constructs *cLuc-TdNLR1_CC* and *nLuc-TdNLR1_CC* to test protein self-interaction. The recombinant constructs were coexpressed in 4-week-old *N. benthamiana* leaves via infiltration with *A. tumefaciens* strain GV3101 (pSoup-p19). Luciferase activity was detected in coinfiltrated leaves at 48 h after infiltration using a multifunctional imaging system (ChemiDoc). Details of this assay were described previously[66].

Agronomic traits
The introgression line CM42 + *YrTD121* (Chuanmai 42*4/*YrTD121* F_5) and the recurrent parent were grown in the field in the 2023–2024 wheat growing season. Individual plants were spaced 10 cm apart in 1.5-m rows, with 30 cm between rows. CM42 + *YrTD121* and CM42 were sown in 4-row plots and assessed for their response to a mixture of *Pst* races (CYR32, CYR33, CYR34, Zhong4 and HY46-8). To measure agronomic traits, CM42 + *YrTD121* and CM42 were planted in 10-row plots and a range of agronomic traits were compared (Fig. 6). Field management, including weeding, disease and pest control, fertilization and irrigation, followed local standard practices.

Primers
The primers used in this study are listed in Supplementary Table 2.

Statistical analysis
Statistical analysis and data visualization were performed using SPSS 25.0 (IBM Corporation) and Prism v.9.0. Significant differences between the two groups were estimated using a two-tailed Student's *t*-test. *P* values and sample sizes (*n*) are indicated in the figures or figure legends.

Inclusion and ethics
This study's data exclusively come from wheat, without involving any animal experiments. Transgenic planting is subject to strict regulation. All experimental data supporting the findings of this study are included in the 'Data availability'.

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
Data supporting the findings of this work are available in the article and its supplementary information. The wild emmer reference genome assembly WEW_v2.1 was collected from the NCBI database under BioProject no. PRJNA310175 (ref. 52). The transcriptome raw data are available under BioProject no. PRJNA1221773. The genomic sequence of the ptg005881l contig for the *YrTD121* region is available under accession no. PV110151.1. The genomic DNA sequences of *TdNLR1* and *TdNLR2* from TD121 are accessible under GenBank accession nos. OP948046.1 and PP874936.1. The genomic DNA sequences of other *TdNLR* haplotypes in this study are available under accession nos. OP948047.1–OP948060.1, accession no. OR472555.1 for *TdNLR1*, and accession nos. PP874937.1–PP874945.1 and PQ412542.1 for *TdNLR2*. Source data are provided with this paper.

Code availability
All software packages used in the study are publicly available and as described in Methods and Reporting Summary.

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Author contributions

B.H.W., Z.Y.L., T.F., D.C.L. and L.H. conceived and designed the research. Y.L.H., L.H.F. and L.H. developed the mapping population. Y.L.H., Y.Q.L., S.Z.N., M.H. and L.Q.Z. contributed to the development of the mutant lines. Y.L.H., Y.Q.L. and R.J.X. conducted the VIGS. Y.L.H., M.M.L., L.L.D. and R.J.X. performed the gene cloning and expression analysis. K.W. performed the gene editing work. M.M.L. and Z.Y.L. performed some of the transgenic experiments. Y.L.H., M.M.L., L.H.F., B.J., Z.W.Y. and M.H. performed the field stripe rust tests and haplotype analysis of NLRs. Y.L.H., Y.Q.L. and Y.Y.H. contributed to the cell death assay and protein–protein interaction experiments. Y.L.H., M.M.L., S.S.C., B.Y.X. and C.J.X. performed the stripe rust tests in the growth chambers. F.N. and Y.F.J. performed the PacBio sequence assembly and transcriptome analysis. Y.L.H. and Y.H.L. contributed to the phylogenetic analysis. Y.L.H., L.Q.Z. and L.L.D. developed the introgression lines. Y.L.H., B.H.W., Z.Y.L., T.F., D.C.L. and L.H. wrote the paper with input from all coauthors.

Competing interests

L.H., Y.L.H., D.C.L., B.H.W., L.H.F., L.Q.Z., Z.W.Y., S.Z.N. and M.H. declare using the TdNLR sequences under a China patent filing no. 202211502138.2, which are based on the results described. L.H., Y.L.H., D.C.L., B.H.W., L.Q.Z., Z.W.Y., S.Z.N. and M.H. declare competing interest in the use of the markers linked to *YrTD121* under a China patent filing no. 202110932684.9. The other authors declare no competing interests.

Additional information

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Reporting Summary

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	The ImageJ software v1.8.0.345 (https://imagej.net/ij/) was used to quantify the percentage of leaf surface covered with sporulating uredinia. The Epson Expression 11000XL scanner was use to evaluate the grains width and length.
Data analysis	The prediction of protein conserved domains was carried our using online CD-Search interface (https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi) and LRR domain prediction tool (https://lrrpredictor.biochim.ro/). Structure prediction: AlphaFold v3.0. ClustalW in MEGA software (v11) was used for multiple sequence alignments. Phylogenetic tree was visualized using iTOL (https://itol.embl.de/). Statistical analysis and data visualization were performed using SPSS 25.0 (SPSS, Inc., Chicago, IL) and GraphPad Prism v9.0.

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Data supporting the findings of this work are available within the paper and its Supplementary Information.The wild emmer reference genome assembly WEW_v2.1 was collected from National Center for Biotechnology Information (NCBI) database under BioProject PRJNA310175 (ref. 52). Transcriptome raw data for this project are available as BioProject PRJNA1221773. The genomic sequence of ptg005881l contig for YrTD121 region is available as accession number PV110151. The genomic DNA sequences of TdNLR1 and TdNLR2 from TD121 are accessible under NCBI GenBank accessions OP948046 and PP874936. The genomic DNA sequences of other TdNLR haplotypes in this project are available as accession numbers OP948047–OP948060, OR472555 for TdNLR1; and PP874937–PP874945, PQ412542 for TdNLR2. Source data are provided with this paper.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N/A
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Population characteristics	N/A
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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The sample size for all experiments was determined based on sample availability and the requirements of statistical analysis. 145 F2 plants of LDN × TD121 were used to genetic analysis and preliminary mapping of YrTD121. A total of 5,256 F2 plants of LDN × TD121 were further used to construct a high-resolution genetic map of YrTD121. The population size is bigger than most population sizes in similar studies (Periyannan et al. 2013, Science, 341,786-788; Saintenac et al. 2013, Science, 341, 783-786; Fu et al. 2009, Science, 323, 1357-1359; Li et al. 2023, Nat. Commun., 14, 6072).
Data exclusions	No data was excluded from analyses.
Replication	All experiments were repeated at least three times and multiple biological replicates were used in all experiments as indicated. Conclusions drawn in this study are derived from at least three consistent independent experiments.
Randomization	We consistently selected samples at random according to genotype for our experiments. EMS populations and controls were randomly selected for phenotyping as normal practice to identify susceptible mutants. Homozygous recombinant lines and transgenic plants were grown randomly with in the same experiment plot to minimize unexpected environmental variation during experimentation. It is not relevant to other experiments as no subjects were allocated to experimental groups.
Blinding	Blinding was performed when phenotyping the mutant lines/transgenic plants, gene expression analysis. It is not performed when carrying out phylogenetic analysis, gene haplotype sequencing analysis, protein structure analysis, and functional validation as required by the experiments.

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We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

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☒	☐ Dual use research of concern
☐	☒ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Commercial antibodies: anti-GFP (Sangon Biotech, Shanghai, D110008; 1:5000 dilution) and HRP-conjugated goat anti-mouse IgG (Sangon Biotech, D110087; 1:10000 dilution).
Validation	Anti-GFP and goat anti-mouse IgG were used in our immunoblot analysis in Nicotiana benthamiana expressing TdNLR1-YFP, TdNLR1_CC-YFP, TdNLR1_CCdel-YFP, TdNLR1_CCh4-YFP, TdNLR1_NB-YFP, TdNLR1_LRR-YFP, TdNLR1_CC-NB-YFP, TdNLR1_NB-LRR-YFP, TdNLR2_Un-YFP, TdNLR2_NB-YFP, TdNLR2_LRR-YFP, TdNLR2_Un-NB-YFP, TdNLR2_NB-LRR-YFP, and TdNLR2-YFP. The results showed that the antibody allow detection of the specific corresponding protein with expected size.

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☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	The wild emmer wheat accessions used in this study were kept in Wild Cereal Gene Bank, the University of Haifa, Israel. The genetically modified seeds and YrTD121 introgression lines are produced by Triticeae Research Institute, Sichuan Agricultural University and preserved in our lab.
Novel plant genotypes	The TdNLR1 and/or TdNLR2 transgenic plants were generated in common wheat cultivar Fielder using the PureWheat technique. Positive transgenic plants were identified and self-pollinated. For functional verification, three(TdNLR1+TdNLR2), two(TdNLR1), two (TdNLR2) independent T1 complementary transgenic lines were analyzed. Three(TdNLR1+TdNLR2), three(TdNLR1), three (TdNLR2) overexperssion T1 lines were analyzed. For CRISPR/Cas9-mediated gene editing, the guide RNA (ACGTCAGCGACCGGCGCACCAGG/ TACGGGGTCGAGAACCTTACAG target TdNLR1; ATGCCCTGGCTGAACGTACAGGG/AGATACAGGCATCTTCTGCTGGG target TdNLR2) targets the exons, and three (TdNLR1) and two (TdNLR2) independent lines were produced. Primers are listed in the Supplementary Table 3 file.
Authentication	Gene and vector specific primers are paired to detect presence of the exogenous vectors in the transgenic lines. PCR-based sequencing was used to identify mutations in the target genes.